

CYBERSECURITY TOOL LEARNING PORTFOLIO

VirusTotal

Indicator lookup, multi-engine analysis, report review, and sandbox behaviour

Course Learning Portfolio

A focused record of the VirusTotal features I learned and how I used them during a SOC investigation.

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Tool	VirusTotal
Learning area	Threat intelligence and indicator analysis
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Portfolio note

This portfolio describes only the functions and report areas I used in VirusTotal.

1. Tool Overview

VirusTotal is an online security analysis service that brings together results from multiple antivirus engines, URL scanners, file characterization tools, sandbox systems, and community contributions. I used it to look up indicators and review existing reports for files, URLs, domains, and IP addresses.

The tool does not create one independent malware verdict of its own. Instead, it presents the results returned by different security vendors and supporting analysis systems. This means I must interpret the report carefully rather than treating one detection as automatic proof of malicious activity.

What I learned to use

- Search for a file by its MD5, SHA-1, or SHA-256 hash.
- Submit a file for analysis when an existing report is not available and the file is safe to share.
- Search for or analyse a complete URL beginning with http or https.
- Look up domain and IP address reports.
- Read detection results from individual security vendors.
- Review file details, relationships, sandbox behaviour, and community information when available.
- Use the report as supporting evidence during alert investigation.

Main VirusTotal functions

Feature	Purpose
File analysis	Checks a submitted file with multiple analysis engines and displays a combined report.
URL analysis	Checks a submitted URL and displays how URL-scanning partners classify it.
Hash search	Retrieves the latest stored report using an MD5, SHA-1, or SHA-256 value.
Domain lookup	Shows available security and relationship information for a domain.
IP address lookup	Shows available security and passive-DNS-related information for an IP address.
Report tabs	Organise detections, technical details, relationships, behaviour, and community information.



VirusTotal search interface used to submit or look up an indicator.

2. Searching and Submitting Indicators

2.1 Searching by file hash

A file hash is a unique value calculated from the contents of a file. VirusTotal accepts MD5, SHA-1, and SHA-256 hashes in the search box. Searching by hash is useful because I can review an existing report without uploading the file again.

1. Copy the file hash from the SIEM, EDR, email investigation, or endpoint record.
2. Paste the hash into the VirusTotal search bar.
3. Open the latest available report.
4. Confirm that the hash in the report matches the indicator from the alert.
5. Review detection, file details, relationships, behaviour, and community information.

2.2 Submitting a file

When a report is not already available, VirusTotal can accept a file for analysis. The uploaded file is processed by participating security engines and other analysis tools, and a report is created when the analysis finishes. I upload the file from malware bazaar. I did not open or execute the sample on my normal personal computer.

Important upload rule

I should not upload confidential, private, customer, or company files to the standard VirusTotal service. Standard submissions may be shared within the VirusTotal ecosystem. When I already have a file hash, I should search the hash first instead of uploading the file.

13/60 security vendors flagged this file as malicious

Reanalyze Similar More

a92b69a3b80cb026568a05ce6fd025a06c02d791044c427b847c75cc6bbac2a5
a92b69a3b80cb026568a05ce6fd025a06c02d791044c427b847c75cc6bbac2a5.js

Size 392.11 KB Last Analysis Date a moment ago JS

Community Score 13 / 60 -11

DETECTION DETAILS BEHAVIOR COMMUNITY 3

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to [automate checks](#).

Popular threat label trojan.asynrat/drop Threat categories trojan dropper Family labels asynrat drop sagent

Security vendors' analysis Do you want to automate check

AliCloud Trojan[dropper]:Javascript/AsyncRAT.SB... Avast JS:Agent-EOT [Trj]

VirusTotal search result showing the indicator and the report opened for review.

2.3 Searching or analyzing a URL

VirusTotal accepts a full URL beginning with http or https. It normalizes the URL, checks whether a report already exists, and returns the most recent available analysis. The report shows how participating URL scanners classify the address. The url which I upload is `https://tuasesoriadigital[.]es/default[.]dat` from malware bazar.

The screenshot shows the VirusTotal interface with the URL `https://tuasesoriadigital.es/default.dat` in the search bar. The 'DETECTION' tab is active. A banner at the top encourages joining the community. Below, the 'Security vendors' analysis' section displays a table of results:

Vendor	Detection	Vendor	Detection
ADMINUSLabs	Malicious	BitDefender	Malware
Certego	Malicious	Chong Lua Dao	Malicious
CRDF	Malicious	CyRadar	Malware
Emsisoft	Malware	ESET	Malware
Fortinet	Malware	G-Data	Malware
Gridinsoft	Phishing	Kaspersky	Malware

Search result showing the indicator and the report opened for review.

2.4 Looking up a domain or IP address

A domain or IP address can be entered directly into the search box. VirusTotal then displays available information and related activity, such as connected URLs, files, or network infrastructure recorded in its datasets. The amount of visible relationship data can depend on the report and account access. I used ip address of 141[.]105[.]71[.]82 on virustotal to show what results are shown when we scan ip on the virustotal.

The screenshot shows the VirusTotal interface with the IP address `141.105.71.82` in the search bar. The 'DETECTION' tab is active. A summary card at the top left shows a 'Community Score' of 10/91. The main card displays the IP address and its associated information:

- 141.105.71.82 (141.105.64.0/21)
- AS 50867 (Hostkey B.v.)
- RU (Russia)
- Last Analysis Date: 1 day ago

Below the summary card, the 'DETECTION' tab shows a banner for the community and a section for 'Security vendors' analysis' with the following results:

Vendor	Detection	Vendor	Detection
ADMINUSLabs	Malicious	alphaMountain.ai	Malicious

Search result showing the indicator and the report opened for review.

3. Understanding a File Report

A VirusTotal file report is divided into sections. Each section answers a different investigation question. I learned to move through the report in a fixed order instead of looking only at the detection number.

3.1 Summary and detection result

The report summary shows how many security engines classified the file as malicious or suspicious compared with the total number of engines that produced a result. It also displays the file identity and analysis time.

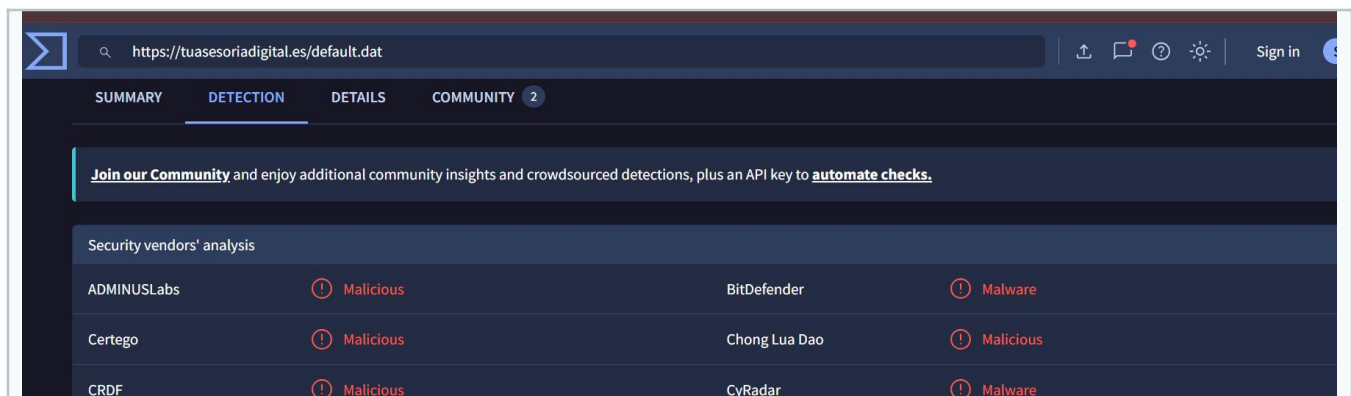
- Detection count or ratio
- File name or meaningful name
- File type
- File hashes
- Analysis date
- Community score, when available

3.2 Detection tab

The Detection tab lists the participating security engines and the result returned by each one. An engine may label the file as malicious, suspicious, undetected, unsupported, or unable to complete the analysis. When a vendor detects the file, VirusTotal also displays that vendor's detection name. Use `https://tuasesoriadigital[.]es/default[.]dat` **from** malware bazar.

How I interpret detections

I compare the number of detection, the names used by different engines, and the rest of the report. One detection can be a false positive, while several consistent detection provide stronger evidence. VirusTotal is supporting evidence, not the only evidence in the case.



Multi-engine detection results for the file hash/url being investigated.

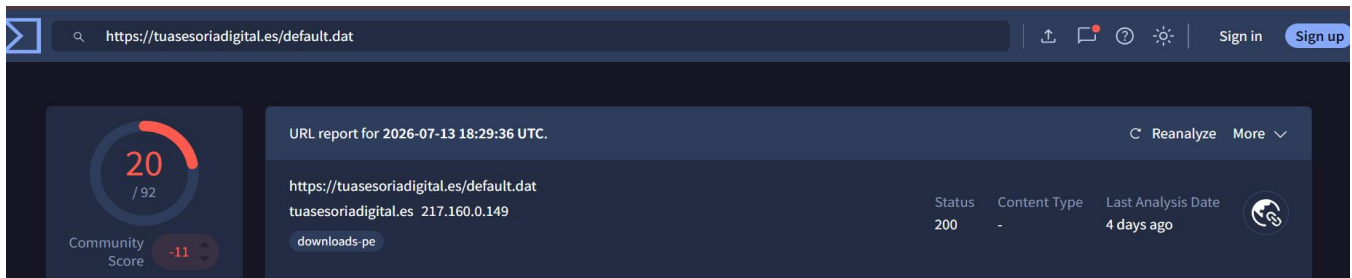
3.3 Details tab

The Details tab contains technical information that helps identify and describe the file. The exact fields vary by file type, but the report can include:

- MD5, SHA-1, and SHA-256 hashes
- File type and size
- Known file names
- First submission and last submission dates
- Last analysis date

- File-format-specific metadata
- Digital-signature or certificate information when available

Used `hxxps://tuasesoriadigital[.]es/default[.]dat` **from** malware bazar.



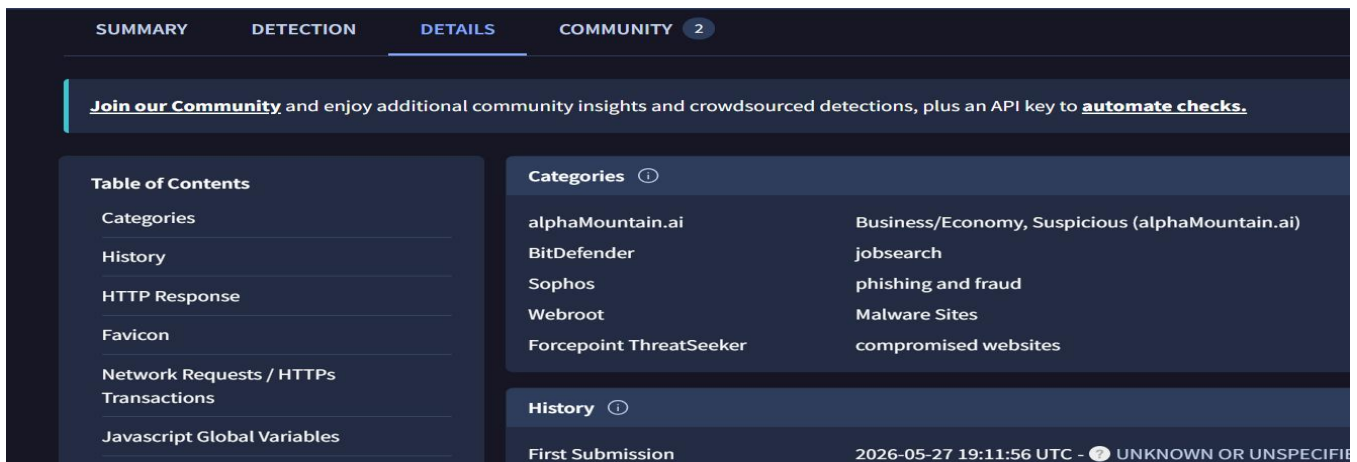
URL report for 2026-07-13 18:29:36 UTC. Reanalyze More

`https://tuasesoriadigital.es/default.dat`
 tuasesoriadigital.es 217.160.0.149

Status: 200 Content Type: - Last Analysis Date: 4 days ago

Community Score: 20 / 92 -11

downloads-pe



SUMMARY **DETECTION** **DETAILS** **COMMUNITY** 2

[Join our Community](#) and enjoy additional community insights and crowdsourced detections, plus an API key to [automate checks](#).

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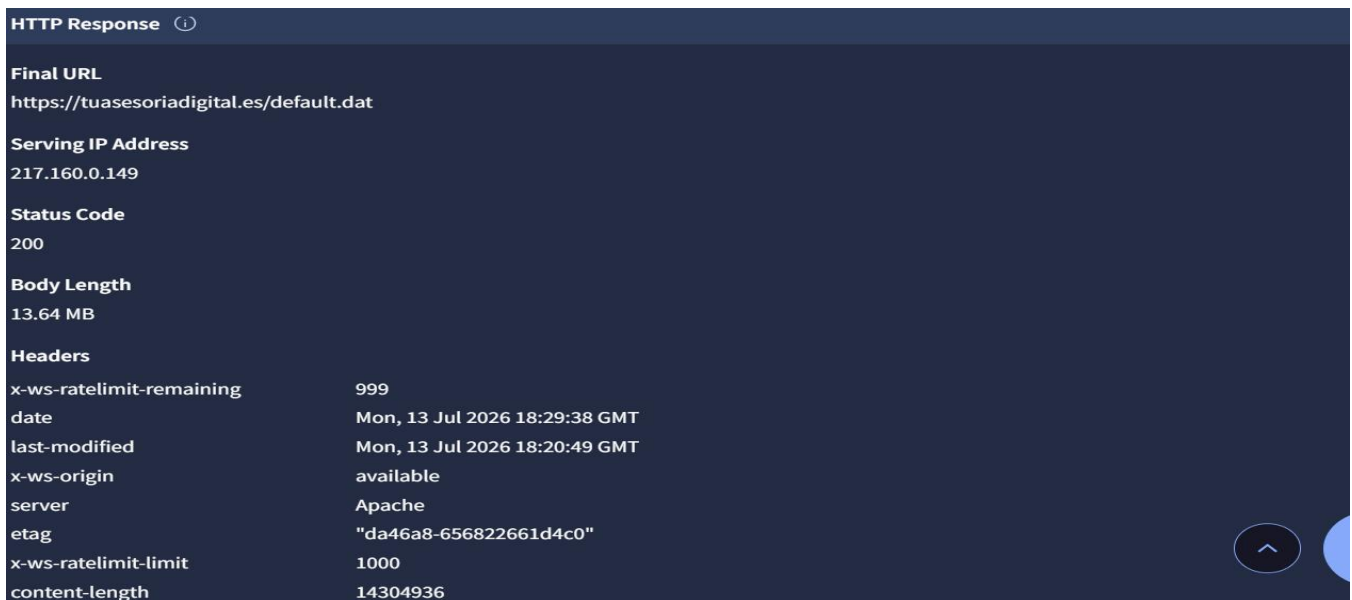
- Categories
- History
- HTTP Response
- Favicon
- Network Requests / HTTPs Transactions
- Javascript Global Variables

Categories ⓘ

alphaMountain.ai	Business/Economy, Suspicious (alphaMountain.ai)
BitDefender	jobsearch
Sophos	phishing and fraud
Webroot	Malware Sites
Forcepoint ThreatSeeker	compromised websites

History ⓘ

First Submission	2026-05-27 19:11:56 UTC - UNKNOWN OR UNSPECIFIED
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HTTP Response ⓘ

Final URL
`https://tuasesoriadigital.es/default.dat`

Serving IP Address
 217.160.0.149

Status Code
 200

Body Length
 13.64 MB

Headers

x-ws-ratelimit-remaining	999
date	Mon, 13 Jul 2026 18:29:38 GMT
last-modified	Mon, 13 Jul 2026 18:20:49 GMT
x-ws-origin	available
server	Apache
etag	"da46a8-656822661d4c0"
x-ws-ratelimit-limit	1000
content-length	14304936

Favicon ⓘ

dhash:

699e96b6256dc869

raw md5:

7661af74b7f80e629a71cdf0be1a8067

Network Requests / HTTPs Transactions (1/1) ⓘ

URL	Headers	GTI Score	Status	Detections ⓘ	Relation Date ⓘ
https://tuasesoriadigital.es/default.dat downloads-pe	9 ⓘ	-	200	20 / 92	2026-07-13 18:29:36 UTC

File metadata and hashes/url used to confirm the identity of the analyzed file.

4. Relations, Behaviour

4.1 Relations tab

The Relations area shows links between the current object and other objects in VirusTotal. For a file, this may include URLs connected with the file, files that contain it, files it contains, related domains, or related IP addresses. Relationship availability varies by object and account access.

I use this section to extend the investigation beyond the original hash. A related URL, domain, or IP address can become a new indicator to search in SIEM, EDR, DNS, proxy, or firewall logs. Used IP address of virus total to show relations.

The screenshot shows the VirusTotal interface for the IP address 141.105.71.82. The top navigation bar includes a search bar with the IP address, a 'Sign in' button, and a 'Sign up' button. The main header displays a circular security score of 10/91, a warning that 10/91 security vendors flagged this IP as malicious, and the IP address 141.105.71.82 (141.105.64.0/21) with AS 50867 (Hostkey B.V.) and a Russian flag. The 'RELATIONS' tab is selected, showing a table of Passive DNS Replication data.

Date resolved	Detections	Resolver	Domain
2026-05-08	0 / 91	VirusTotal	api.node-core.net
2025-01-24	0 / 91	VirusTotal	net.cosmonet.online
2024-03-11	0 / 91	VirusTotal	46252.hostkey.in

Related files, URLs, domains, or IP addresses connected with the analyzed item.

The screenshot shows the VirusTotal interface for the IP address 141.105.71.82, specifically the 'Communicating Files' tab. It displays a table of files that have communicated with the analyzed IP address.

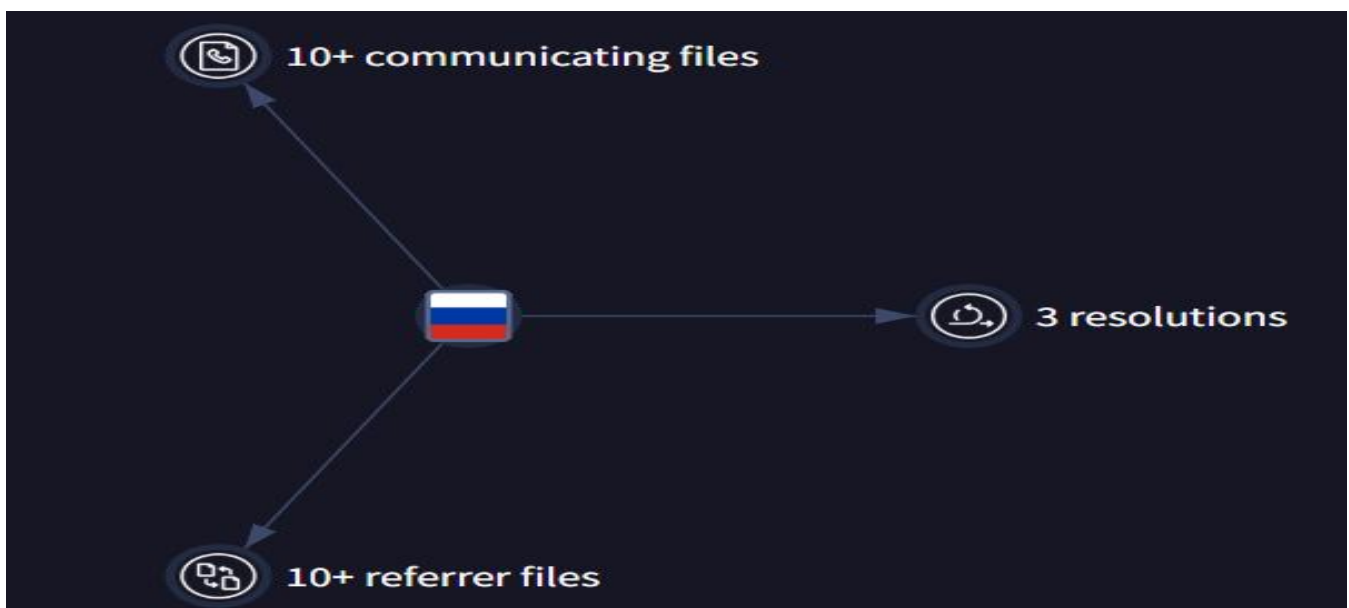
Scanned	Detections	Type	Name
2021-04-03	39 / 69	Win32 EXE	0b93a448917f6e07fb66571ae9987adcaa812969063470bb1c696fe3626fb181
2021-04-21	43 / 69	Win32 EXE	жирнейшие почты с аккаунтами steam и вк валид 70% много GTA 5 Social Club.exe
2025-03-22	53 / 73	Win32 EXE	705ccde0c3d32649b4ae8dbe6a574a91.virus
2021-09-25	40 / 69	Win32 EXE	2ed84f3c5769bbd026b937d7e6423663c65da54a114e7784891c101e20583505
2020-05-12	47 / 73	Win32 EXE	df4036d08725e8f5144cc82889e42b09.virus
2026-07-07	55 / 68	Win32 EXE	sample_5
2021-01-02	50 / 71	Win32 EXE	USERDATA.EXE
2021-01-10	37 / 68	Win32 EXE	eYwixLeLbiHtzyoBeUJqh.exe
2020-12-28	38 / 71	Win32 EXE	USERDATA.EXE
2021-04-02	35 / 70	Win32 EXE	qsfMdkRCgKhlcxHDvmglyqjyIZgD.exe

Related files, URLs, domains, or IP addresses connected with the analyzed item.

Files Referring (59) ⓘ

Scanned	Detections	Type	Name
2026-07-07	55 / 68	Win32 EXE	sample_5
2026-07-06	50 / 66	Win32 EXE	Stealer_Build.exe
2026-07-06	54 / 69	Win32 EXE	z668 1.6.exe
2025-03-22	53 / 73	Win32 EXE	705ccde0c3d32649b4ae8dbe6a574a91.virus
2025-03-15	49 / 73	Win32 EXE	mtstocom.exe
2025-03-09	52 / 72	Win32 EXE	2923f0034e9de3191745e18429db23dfd44dd19e7b4fdc9e0c98c057f192f38f
2025-03-07	51 / 71	Win32 EXE	e6be47d6b3d680497a0bc9cf30bd4278.virus
2025-01-28	44 / 72	Win32 EXE	4ea0d16f5202b5c7c787bf5db608a6a612655c8006908f5b688aeceb22b2f281
2025-01-27	51 / 72	Win32 EXE	7307083
2025-01-23	52 / 72	Win32 EXE	ExtreamHackV2.1.3.exe

Related files, URLs, domains, or IP addresses connected with the analyzed item.



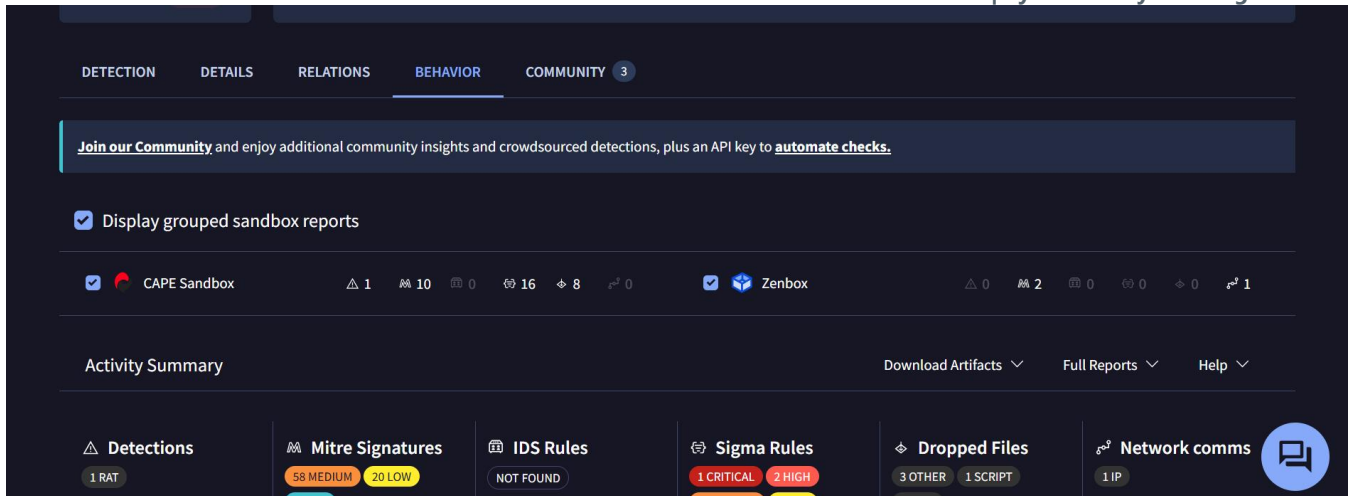
Graph Summary

4.2 Behaviour tab

The Behaviour area summarizes actions observed when a file is executed or opened in integrated sandbox environments. The available information depends on the file and completed sandbox analyses.

- Processes and command executions
- Files opened, written, deleted, or modified
- Services created, started, stopped, or deleted
- Network connections and IDS alerts
- Process injection or terminated processes when observed
- MITRE ATT&CK techniques linked to observed behaviour when available

This tab is useful because it shows what the file attempted to do, not only what antivirus engines called it. I compare the observed actions with endpoint evidence from the actual affected device.



Sandbox behaviour showing processes, commands, file activity, or network activity.

5. Understanding URL, Domain, and IP Reports

5.1 URL report

A URL report shows how participating URL scanners classify the submitted address. Possible labels can include harmless or clean, unrated, malware, phishing, malicious, suspicious, or spam, depending on the reviewing vendor.

- Number of vendors that consider the URL harmful
- Normalized URL
- Related domain
- Review date and time
- Content type and available HTTP information
- Individual vendor classifications
- Community reputation and comments
- Related objects or graph options when available

5.2 Domain report

A domain report provides security information about the domain and available related activity. I use it to check whether the domain is connected with suspicious URLs, malware samples, IP addresses, or other recorded objects.

5.3 IP address report

An IP report provides available security and passive-DNS-related information for the address. It may show connected domains, URLs, files, or incidents recorded in VirusTotal. I use these relationships to decide what else should be searched in network logs.

Report interpretation

A clean or undetected result does not prove that an indicator is safe. A malicious result also needs context. I use VirusTotal together with the original alert, endpoint data, logs, network evidence, and the timing of the activity.

6. How I Use VirusTotal During an Investigation

I use VirusTotal as an enrichment tool after I collect an indicator from an alert. The report helps me understand how the indicator has been classified and what related information is available.

1. Collect the indicator from the alert, such as a file hash, URL, domain, or IP address.
2. Confirm that the value was copied correctly and note the alert time.
3. Search the indicator in VirusTotal. For files, I search the hash before considering an upload.
4. Check the report date so I know how current the result is.
5. Review the detection summary and individual vendor results.
6. Open the Details tab to confirm hashes, file type, names, and submission information.
7. Review Relations to identify connected files, URLs, domains, or IP addresses when available.
8. Review Behaviour for observed commands, processes, file changes, services, and network activity.
9. Read relevant community comments or votes as supporting context.
10. Compare the VirusTotal findings with SIEM logs, EDR activity, and network evidence.
11. Record the result in the case notes and explain how it affected the final classification.

Example investigation note

Case note example

The SHA-256 hash from the endpoint alert was searched in VirusTotal. Multiple engines classified the file as malicious and used similar Trojan labels. The file details matched the hash and file type recorded in EDR. Sandbox behaviour showed command execution, file creation, and an outbound connection to a related domain. The same domain was then searched in proxy logs. VirusTotal results were used as supporting evidence and the alert was escalated for containment.

What I record from VirusTotal

Information	What I document
Indicator	The exact hash, URL, domain, or IP address searched.
Report date	The analysis or review time shown in the report.
Detection summary	The number and type of vendor detections.
Important labels	Consistent malware, phishing, or suspicious classifications.
Technical details	File hashes, type, names, and relevant metadata.
Related indicators	Any connected URLs, domains, IP addresses, or files used for further searches.
Observed behaviour	Relevant sandbox processes, commands, file changes, or network activity.
Conclusion	How the report supports or does not support the alert.

7. What I Learned

After practising with VirusTotal, I can use the tool to enrich a security alert without relying only on the alert title. I understand the difference between a detection result, technical file details, sandbox behaviour, relationships, and community information.

Learning outcomes

- I can search files using MD5, SHA-1, and SHA-256 hashes.
- I can analyze or look up URLs, domains, and IP addresses.
- I can read the multi-engine detection summary and individual vendor labels.
- I can verify file identity using hashes and metadata in the Details tab.
- I can use Relations to discover connected indicators.
- I can read sandbox behaviour to understand observed file actions.
- I can review community information without treating it as confirmed evidence by itself.
- I can document VirusTotal findings in a SOC investigation.
- I know not to upload confidential files to the standard service.

My main takeaway

Main takeaway

VirusTotal helps me collect threat context quickly, but it does not replace investigation. I must combine its results with the original alert, logs, endpoint activity, network evidence, and business context before deciding whether an event is malicious or benign.

9. VirusTotal Terms Used in This Portfolio

Term	Meaning
Hash	A value calculated from a file and used to identify it. VirusTotal accepts MD5, SHA-1, and SHA-256 for file report searches.
Detection	A classification returned by a participating security engine.
Undetected	The engine did not classify the item as malicious in that analysis.
False positive	A legitimate item incorrectly classified as malicious or suspicious.
Relation	A recorded connection between objects such as files, URLs, domains, and IP addresses.
Sandbox behaviour	Actions observed when a file is executed or opened in a controlled environment.
Community score	A reputation value influenced by votes from registered community members.
Reanalyse	Run or request a newer analysis when the interface provides that option.

Official sources used

- **VirusTotal documentation - How it works** - <https://docs.virustotal.com/docs/how-it-works>
- **VirusTotal documentation - Searching** - <https://docs.virustotal.com/docs/searching>
- **VirusTotal documentation - Reports** - <https://docs.virustotal.com/docs/results-reports>
- **VirusTotal API reference - Files Behaviour** - <https://docs.virustotal.com/reference/file-behaviour-summary>
- **VirusTotal API reference - Relationships** - <https://docs.virustotal.com/reference/relationships>
- **VirusTotal documentation - Community** - <https://docs.virustotal.com/docs/community>

Scope note

This document intentionally focuses on the standard VirusTotal functions used for indicator lookup and report review. It does not describe unrelated malware-analysis tools or general SOC topics.